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Towards Enhanced Attack Detection and Explanation in Intrusion Detection System-based IoT Environment Data

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ABSTRACT Securing the Internet of Things (IoT) against cyber threats is a formidable challenge, and Intrusion Detection Systems (IDS) play a critical role in this effort. However, the lack of transparent explanations for IDS decisions remains a significant concern. In response, we introduce a novel approach that leverages a blending model for attack classification and integrates counterfactual and Local Interpretable Model-Agnostic Explanations (LIME) techniques to enhance explanations. To assess the effectiveness of our approach, we conducted experiments using the recently introduced CICIoT2023 and IoTID20 datasets. These datasets are real-time and large-scale benchmark datasets for IoT environment attacks, offering a realistic and challenging scenario that captures the intricacies of intrusion detection in dynamic IoT environments. Our experimental results demonstrate significant improvements in attack detection accuracy compared to conventional IDS methods. Furthermore, our proposed approach provides clear and interpretable insights into the factors influencing classification decisions, empowering users to make informed security choices. Integrating blending model classification and explanation techniques enhances the security and reliability of IoT systems. Therefore, this work represents a significant advancement in IoT intrusion detection, offering a robust and transparent defense against large-scale cyber-attacks of IoT environment data.

INDEX TERMS Counterfactual, ensemble blending model, IoT, IDS, LIME, XAI.

I. INTRODUCTION

THE rapid growth of the Internet of Things (IoT) has ushered in a new era of interconnected devices, revolutionizing various industries [1]. However, the proliferation of IoT devices has also exposed significant security challenges, as these interconnected systems are vulnerable to diverse forms of attacks [2]. Enhancing attack detection and explanation capabilities is imperative to ensure the security of IoT environments [3].

In the realm of software-defined wireless sensor networks (SDWSN), security and quality of service (QoS) challenges take on paramount significance. SDWSN represents a pivotal evolution in wireless sensor technology, enabling dynamic reconfiguration and optimization of sensor network resources based on the ever-changing demands of IoT applications [4]. This dynamism, while offering unprecedented flexibility, also introduces vulnerabilities. Security threats in SDWSN, such as intrusion attempts and data breaches, can disrupt critical data flows, compromise sensitive information, and impact the integrity of the entire network [5]. Furthermore, ensuring QoS in SDWSN is vital, as it directly impacts the reliability and responsiveness of IoT applications, which often serve critical functions in areas such as healthcare [6], infrastructure monitoring [7], and environmental sensing [8]. Failure to address security and QoS challenges in SDWSN jeopardizes data integrity and network functionality and has broader im-

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plications for the adoption and success of IoT applications in diverse sectors. In this context, our research endeavors to enhance security and QoS, aiming to fortify the foundation of IoT technologies and contribute to realizing their full potential.

The imperative drives this work to bolster IoT security, a challenge accentuated by the rapid proliferation of IoT devices across industries. As IoT ecosystems expand, they become increasingly susceptible to various security threats, including intrusion attempts, data breaches, and service disruptions. Traditional security measures must often catch up in IoT's dynamic and complex environment. It necessitates the development of innovative security solutions that can effectively detect and mitigate attacks and provide transparent explanations for their decisions [9], [10].

As a subset of ensemble learning techniques, the Blending model has demonstrated their prowess in various domains by combining the predictions of multiple models [11]. By aggregating diverse model outputs, blending models can enhance accuracy, robustness, and generalization capabilities [12]. In this paper, we adapt and apply the blending model concept to the domain of IoT security, aiming to improve attack detection accuracy and provide interpretable explanations significantly.

While prior research has shown the potential of ensemble blending model in enhancing security measures [13]–[15], their application to IoT security presents a unique and exciting opportunity [16]. IoT environments are characterized by a dynamic and evolving threat landscape, making them a challenging domain for traditional security methods. Blending models, with their adaptability and ability to leverage the strengths of diverse models, hold promise in bolstering security within IoT ecosystems.

Building upon the foundation laid by previous research, this paper addresses the current challenge by proposing an innovative approach that leverages a blending model for attack classification and incorporates counterfactual [17] and LIME (Local Interpretable Model-Agnostic Explanations) techniques [18] to enhance explanations. Through rigorous experimentation and evaluation on two representative datasets, CICIoT2023 [19] and IoTID20 [20], we demonstrate the superiority of our proposed blending model in terms of attack detection accuracy and explainability. This study responds to the unique security challenges posed by IoT systems, emphasizing the importance of combining advanced detection capabilities with transparent explanations to fortify IoT security effectively.

The main contributions of this paper include:

- We introduce an approach utilizing a blending model to enhance the effectiveness of attack detection and provide transparent explanations in IoT environments. By harnessing the collective intelligence of multiple models, our approach significantly improves both accuracy and interpretability.
- Through an extensive series of experiments conducted on two benchmark datasets, our research underlines the superior performance of the blending model in comparison to baseline methods. We examine key evaluation metrics such as attack detection accuracy, AUC, recall, and F1 score. Additionally, our blending method surpasses other existing methods when applied to the same datasets. This contribution underscores the efficacy of our proposed approach in fortifying IoT security.
- Our proposed model elevates attack detection accuracy and offers interpretable explanations, providing valuable insights into the characteristics and nature of the identified attacks. We conduct a comprehensive analysis of the interpretability and explainability aspects of the model, shedding light on the contributing factors influencing the model's classification results.

The rest of the paper is organized as follows: In Section II, we comprehensively review related work in IoT security, ensemble learning, and Explainable Artificial Intelligence (XAI) techniques. Section III presents the methodology, including a detailed explanation of the blending model and its evaluations. The results and analysis of our experiments are presented in Section IV, highlighting the superior attack detection accuracy achieved by the proposed blending model. Section V discusses the advantages and limitations of the proposed method. Besides, practical considerations and challenges for implementing the proposed model in real-world IoT environments are discussed. Finally, Section VI concludes the paper by summarizing our contributions, discussing the research implications, and outlining future research directions.

II. LITERATURE REVIEW

The rapidly evolving landscape of IoT and its security challenges have led to extensive research in the field of IDS. We have expanded our literature review, focusing on key research works and their contributions to provide a more comprehensive understanding of the existing state-of-the-art solutions.

IDSs are crucial for resource-constrained IoT devices. This research addresses the need for such systems, emphasizing the importance of efficiency and minimal resource consumption in IoT security by Jan et al. [21]. Moreover, Kumar et al. [22] emphasize the importance of designing intelligent cyber attack detection systems for IoT networks. It leverages hybrid feature reduction techniques to enhance the efficiency and accuracy of intrusion detection. The paper discusses the significance of feature reduction in IoT environments with a high volume of data and complex attack patterns. Next, researchers [23] provide the systematic literature review that provides valuable insights into Software-Defined Networking (SDN)-based IoT frameworks. It categorizes existing research and highlights open challenges, shedding light on the role of SDN in shaping the future of IoT. Recently, in [24], the integration of machine learning (ML) into IoT security is pivotal, and this research work explores a framework for detecting Distributed Denial of Service (DDoS) attacks in Software-Defined IoT networks. It underscores the role of ML in enhancing the resilience of IoT systems against evolving attack strategies.

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The field of IoT security has witnessed significant advancements in attack detection techniques. Various approaches have been proposed to identify and mitigate attacks in IoT environments, including anomaly detection, signature-based detection, and ML-based detection. Anomaly detection methods aim to identify deviations from normal behavior, while signature-based methods rely on pre-defined patterns of known attacks. ML-based approaches leverage data-driven models to detect and classify attacks based on learned patterns and features [25], [26].

Despite the progress in attack detection techniques for IoT environments, several limitations persist. Anomaly detection methods often need help with high false-positive rates and face challenges in defining accurate baselines of normal behavior in dynamic IoT environments. Signature-based methods cannot detect novel attacks without pre-defined signatures. ML-based approaches face issues related to the availability of labeled training data, feature selection, and vulnerability to adversarial attacks [27], [28].

Another critical challenge in attack detection is the need for more transparent explanations for the decisions made by intrusion detection systems. Many ML models are considered black boxes, hindering their interpretability. This lack of transparency makes it difficult for security analysts and stakeholders to understand the factors influencing classification outcomes. In security-sensitive environments like IoT, understanding why a particular instance is classified as an attack or benign is crucial for informed decision-making and timely response [18], [29], [30]. Recent study [31] has highlighted the potential and necessity to develop and apply XAI techniques, including in the privacy and security fields.

Ensemble learning, a powerful technique that combines predictions from multiple models, has shown promise in addressing the limitations of individual models. It leverages the collective intelligence of diverse models, resulting in better generalization, increased accuracy, and enhanced robustness. Ensemble learning can mitigate bias, reduce variance, and improve the overall decision-making process [32]–[34]. Additionally, ensemble methods offer opportunities for interpretability and explainability, enabling insights into the decision-making process and providing a comprehensive understanding of complex phenomena [35], [36].

Ensemble blending models, including cybersecurity and intrusion detection, have gained particular attention in security applications. These models combine multiple base models using blending techniques, aggregating the predictions from models with different architectures or algorithms to create an ensemble prediction. By leveraging the strengths of diverse models, ensemble blending models are well-suited to address the challenges posed by IoT security. Moreover, combining the ensemble blending model with interpretability and explainability techniques enhances its transparency, making it easier for security analysts and stakeholders to comprehend the factors influencing classification outcomes. Previous studies have demonstrated the effectiveness of the ensemble blending model in improving detection accuracy, reducing false positives, and providing enhanced explainability and interpretability in security domains [37]–[39]. Besides, Lei et al. [40] utilized an ensemble blending model to enhance anomaly detection in network traffic analysis. Khaledian et al. [41] employed an ensemble blending model for malware detection in computer systems. These studies highlight the potential of the ensemble blending model in security applications and emphasize its advantages in terms of detection accuracy and interpretability.

This paper proposes an innovative approach that employs a blending model for attack classification in IoT environments to address the challenges in IoT security and enhance attack detection. The proposed approach enhances the model's interpretability by integrating counterfactual and LIME techniques, providing transparent and comprehensive explanations for its classification decisions. Rigorous experimentation and evaluation on two representative benchmark datasets demonstrate the superiority of the proposed blending model in terms of attack detection accuracy and explainability.

III. METHODOLOGY

In this section, we provide a comprehensive method that contains four components: data processing and selection, blending model, model evaluation, and classification explanation. The proposed architecture is shown in Figure 1. Besides, we present the pseudo-code of our method in Algorithm 1. The notations are defined in Table 1.

Algorithm 1 Classification and explanation of the proposed method

Input: $\mathcal{DS}_{tr}, \mathcal{DS}_{va}, \mathcal{DS}_{te}, BC$, where $\mathcal{DS} = \{X, Y\}$
Output: EBM , LC , CM, C, L

1: **while** $(i \leq 2)$ $BC_i \leftarrow \texttt{train}(\mathcal{DS}_{tr})$ **end while**
2: $\mathcal{DS}_{ntr} \leftarrow \{Pr, Y\}$, where $Pr_i \leftarrow BC_i(\mathcal{DS}_{va})$
3: $VC \leftarrow \texttt{train}(\mathcal{DS}_{ntr})$
4: $EBM(X) \leftarrow VC(BC_1(X), BC_2(X), BC_3(X))$
5: $LC \leftarrow \texttt{Learning Curve}(EBM(\mathcal{DS}_{te}))$
6: $PR \leftarrow \texttt{Confusion Matrix}(CM(\mathcal{DS}_{te}))$
7: $C \leftarrow \texttt{Counterfactal}(EBM(\mathcal{DS}^j_{te}))$
8: $L \leftarrow \texttt{LIME}(EBM(\mathcal{DS}^j_{te}))$
9: **return** EBM, LC, CM, C, L

In the initial data preprocessing phase, we address the presence of categorical data in the output features of both datasets, CICIoT2023 and IoTID20. We employ label encoding to make this data compatible with ML algorithms, which convert categorical data into a numerical format. This technique involves assigning a unique integer value to each category within the feature, effectively replacing category labels with numerical representations.

Additionally, considering the large number of features in our dataset, 47 for CICIoT2023 and 83 for IoTID20, we select the most crucial features for our model's learning process. We employ the Mean Decrease in Impurity (MDI) method, a widely used and popular technique in prior research [42] [43]. MDI is especially useful for feature selection in decision tree-

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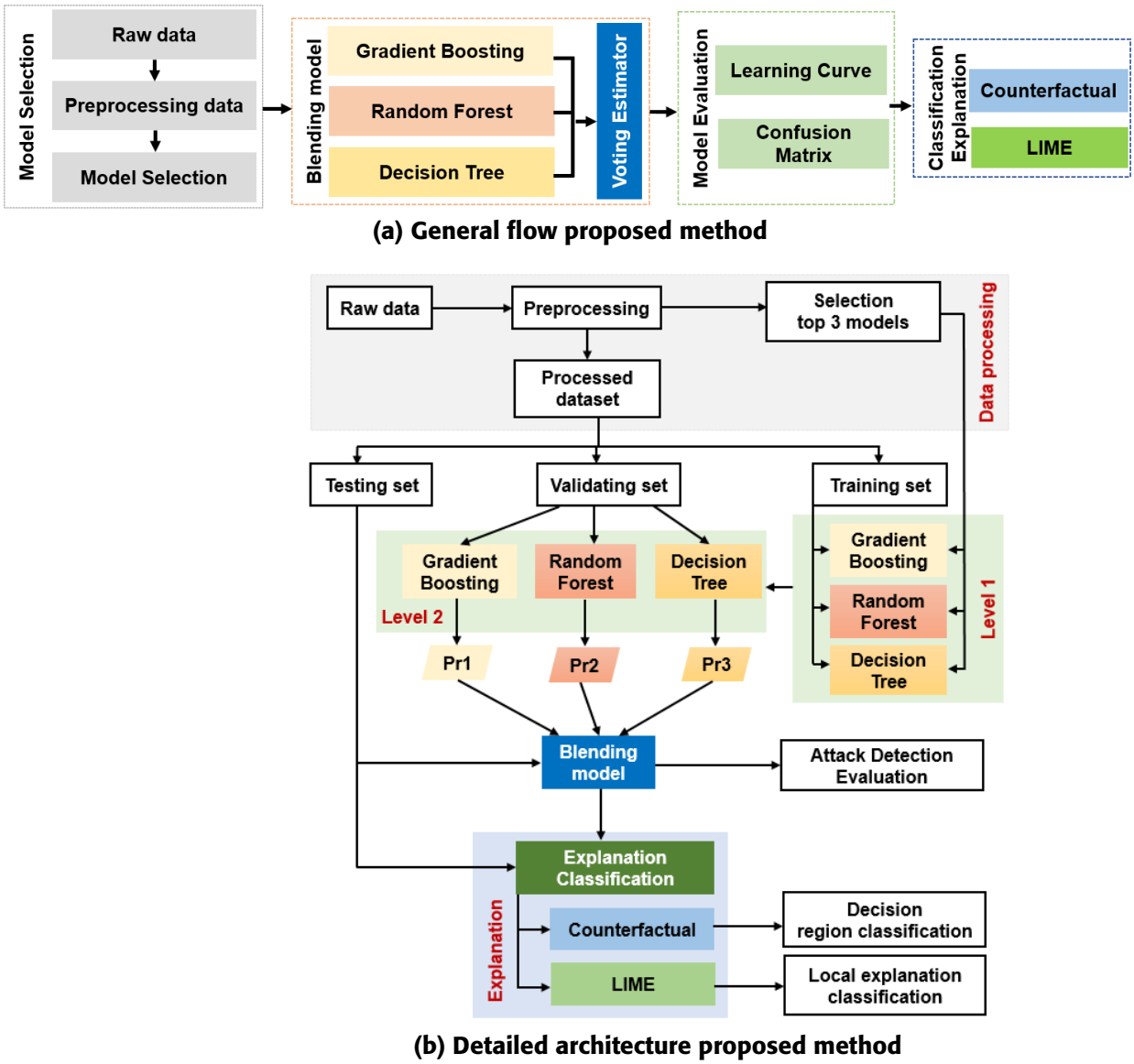

FIGURE 1. The proposed blending and explanation model architecture.

based models like Random Forest. It evaluates the importance of each feature by measuring how much the feature reduces impurity when used in a decision tree split. This reduction in impurity is determined by comparing the impurity of the node before the split to the impurity of the child nodes after the split.

MDI quantifies each feature's contribution to impurity reduction during the decision tree-splitting process. Features with higher MDI values are considered more significant and are prioritized for predictive accuracy, as they play a crucial role in the model's decision-making.

Specifically, we deal with a categorical output feature named 'label' in the CICIoT2023 dataset, which encompasses eight distinct categories: ['BruteForce,' 'DDoS,' 'DoS,' 'Mirai,' 'Normal,' 'Recon,' 'Spoofing,' 'Web-Based']. In the IoTID20 dataset, the categorical output feature, 'Sub Cat,' consists of five distinct categories: ['Mirai,' 'DoS,' 'Scan,' 'Normal,' 'MITM ARP Spoofing']. The label encoding process unfolds in the following steps: (1) Assigning unique labels: Initially, each category in 'label' is assigned a unique integer label ranging from 0 to 7, and 'Sub Cat' receives unique labels ranging from 0 to 4. (2) Data transformation: Subsequently, the original categorical labels within the feature are replaced with corresponding numerical labels, resulting in a transformed dataset ready for ML analysis.

Next, we perform model selection based on processed data. In our experiment, three top high-performance models are selected, including gradient boosting, random forest, and decision tree. We determine these ensemble models as baseline models for the blending model.

In the second component, we delve into ensemble learning techniques, focusing on blending methods. Blending is an

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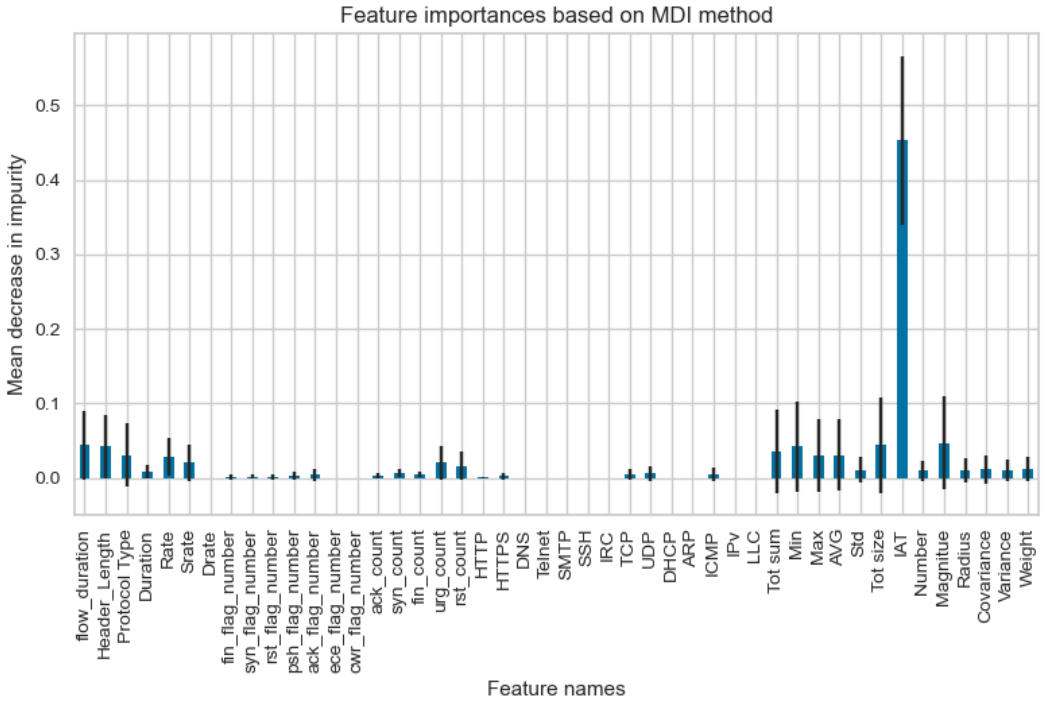
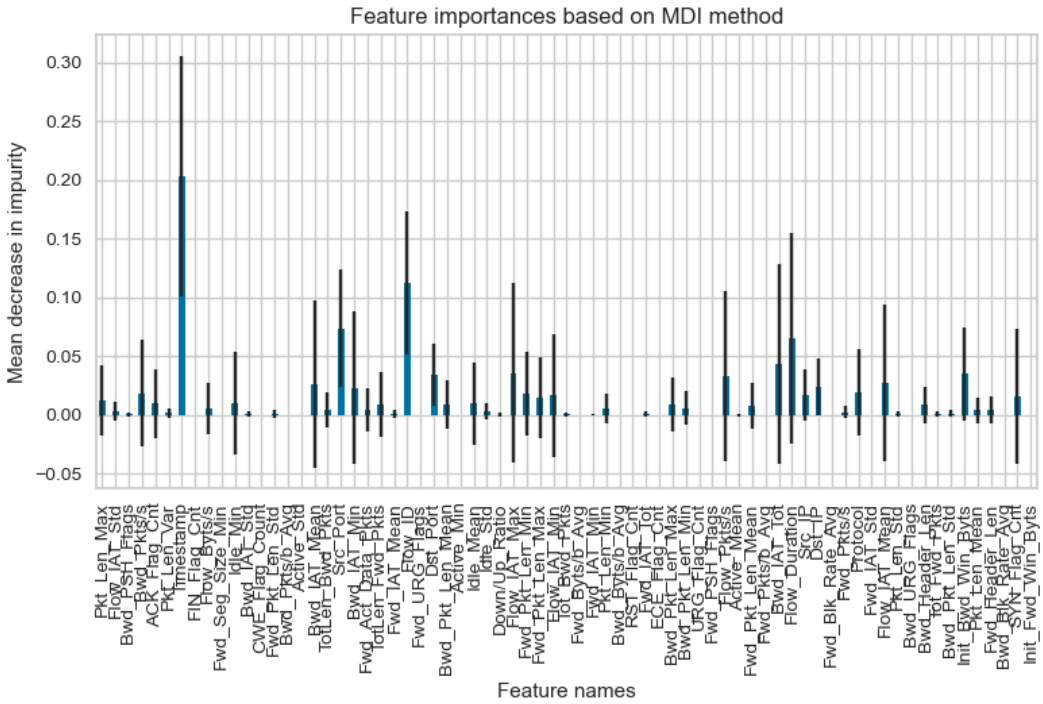


(a)



(b)

FIGURE 2. **Feature important distribution using MDI method. (a) CICIoT2023. (b) IoTID20.**

ensemble technique that uses multiple ML algorithms for training on the same dataset. Unlike other ensemble techniques, there are two levels of ML algorithms used, one is base models, and the second level algorithms are known as meta-models, as shown in detail in Figure 1b. Given: X_train: Training data features; y_train: Training data labels; ; X_val: Validation data features (or X_test for test data); y_val: Validation data labels (or y_test for test data).

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TABLE 1. List of notations used in the paper

Notation	Description
$\mathcal{DS}$	Dataset
$\mathcal{DS}_{tr}$	Training set of the baseline classifiers at level 1
$\mathcal{DS}_{va}$	Validating set of the baseline classifiers at level 2
$\mathcal{DS}_{te}$	Testing set of the baseline classifiers at level 2
X	Set of input features
Y	Class label output
EBM	Ensemble blending model
LC	Learning curve measurement
CM	Confusion matrix measurement
C	Counterfactual explanation
L	LIME explanation
BC	Baseline classifiers: gradient boosting(BC_1), random forest(BC_2), decision tree(BC_3)
$\mathcal{DS}_{ntr}$	New trained dataset of classification
Pr	Predicted value of each baseline classifier on validating data at level 1
VC	Voting classifier to train the new trained dataset at level 2

Step 1: Train the Base Models (Level 1)

Let BC_1, BC_2, and BC_3 be the three baseline models (e.g., gradient boosting, decision tree, and random forest). Train each model on the training data (X_train, y_train) to obtain the models' predictions for the training set:

$$Pred1_train = BC_1(X_train) \tag{1}$$

$$Pred2_train = BC_2(X_train) \tag{2}$$

$$Pred3_train = BC_3(X_train) \tag{3}$$

Step 2: Make Predictions (Level 1): Use the trained models to make predictions on the validation (or test) data (X_val):

$$Pred1_val = BC_1(X_val) \tag{4}$$

$$Pred2_val = BC_2(X_val) \tag{5}$$

$$Pred3_val = BC_3(X_val) \tag{6}$$

Step 3: Create a new training set (Level 2): Concatenate the predictions from the base models (Level 1) to create the new training set for the blending model:

$$\mathcal{DS}_{ntr} = [Pred1_train, Pred2_train, Pred3_train] \tag{7}$$

Step 4: Train the blending model (Level 2): Train the blending model (BM) using the new training set ($\mathcal{DS}_{ntr}$) and the true labels for the validation (or test) set:

$$BM = Train_Model(\mathcal{DS}_{ntr}, y_val) \tag{8}$$

Step 5: Make Final Prediction (FP)(Level 2): Use the trained blending model to make the final prediction on the validation (or test) set:

$$FP = BM([Pred1_val, Pred2_val, Pred3_val]) \tag{9}$$

The blending model works on two levels. In Level 1, the baseline models (BC_1, BC_2, and BC_3) are trained on the training data and used to make predictions on the validation (or test) data. In Level 2, the blending model is trained using the predictions from Level 1 as input features and the true labels of the validation (or test) set. The blending model learns to combine the predictions from the baseline models and create a final prediction that leverages their strengths.

This two-level approach allows the blending model to capture more complex relationships among the base model predictions and improves the overall performance compared to individual models. The final prediction from the blending model can be used for evaluation or making predictions on new, unseen data.

In the third component, we evaluate our blending method in training, validating, and testing data using several evaluation metrics: learning curve and confusion matrix.

- The learning curve, typically represented as a graph, offers valuable insights into an ML model's behavior as it encounters varying amounts of training data. It demonstrates how the model's performance metrics, often denoted as P, evolve concerning the number of training examples indicated as N. Mathematically, this can be expressed as:

$$P = f(N)P = f(N) \tag{10}$$

  where, P represents the model's performance metric, such as accuracy or loss. N signifies the number of training examples.

  Examining the learning curve graph shows how the performance metric P changes concerning N. This visualization aids in understanding critical aspects of the model's behavior, including its capacity to generalize from data and whether it exhibits tendencies like overfitting or underfitting.
- Confusion matrix. The confusion matrix is a fundamental tool for evaluating the performance of classification algorithms. It provides a tabular representation of predicted versus actual class labels. In multi-class classification, there are more than two classes. The confusion matrix extends to cover multiple classes, allowing us to evaluate the model's performance for each class. The confusion matrix for multi-class classification can be mathematically represented as a square matrix with dimensions equal to the number of classes. Each element (i, j) represents the number of instances from class i predicted as class j.

In the final component, the methodology section explains the IoT environment and attack scenarios. We use counterfactual and LIME methods to explain the blending model prediction results.

- Counterfactual: Explanations for region-based models are designed to offer insights into the model's decision-making process by generating instances within a specific region or boundary. Unlike individual instances, these explanations aim to understand the model's decision boundaries for a particular input space region.

  Consider a region-based, like a blending model, which takes a feature vector X as input and produces a binary prediction y_pred (0 or 1). To generate counterfactual

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explanations for a specific region, we follow some processes below:

Define the region: Determine the region of interest in the input space, either based on user-defined criteria or exploration of the model's decision boundaries. Given: X_i: The specific instance we want to generate a counterfactual explanation. Blending model: Our blending model takes a feature vector X as input and produces a binary prediction y_pred (0 or 1).

Choose a target class: Decide which class we want to generate a counterfactual explanation for (0 or 1). We call this the target class T.

Define the optimization problem: Formulate an optimization problem to find a counterfactual instance X_cf that satisfies two conditions: Minimize the distance between X_i and X_cf, denoted as $d(X_i, X_cf)$. For example, this distance can be measured using $L1$ or $L2$ norms. The blending model predicts the target class T for X_cf, i.e., $y_pred(X_cf) = T$.

Mathematically, the optimization problem can be represented as:

$$X_cf = argmin(d(X_i, X_cf)) \tag{11}$$

subject to:

$$y_pred(X_cf) = T \tag{12}$$

Solve the optimization problem: Utilize an optimization algorithm, such as gradient descent, to find the counterfactual instance X_cf that minimizes the distance while satisfying the constraint that the blending model predicts the target class T. The specific optimization algorithm and solver depend on the complexity of the blending model and the chosen distance metric.

Interpret the result: The counterfactual instance X_cf represents a minimal perturbation from the original instance X_i that results in a different prediction (target class T instead of the original prediction). By examining the feature differences between X_i and X_cf, we can understand which features are critical for the blending model's decision and how changes to certain features can alter the prediction outcome.

- LIME: It provides interpretability for black-box models, such as blending models, on a local level. It approximates the black-box model's predictions for a specific instance.

  Given: X_i: The specific instance we want to explain the prediction for. $X_perturbed$: A neighborhood dataset with perturbed instances around X_i. Blending model: Our black-box blending model takes X as input and produces a binary prediction y_pred (0 or 1). Interpretable model: Used to approximate the blending model's decision boundary locally.

  Sample neighborhood data: Generate $X_perturbed$ by sampling similar instances to X_i with random feature perturbations while keeping the label fixed.

  Get predictions for neighborhood data: Use Blending model to predict probabilities for $X_perturbed$, resulting in $y_prob_perturbed$.

  Fit an interpretable model: Train Interpretable model using $X_perturbed$ and the predicted probabilities $y_prob_perturbed$ as labels. The interpretable model approximates the blending model's decision boundary around X_i. For instance, in linear regression, it can be represented as:

$$y_pred_interpretable = w_0 + w_1 \times 1 + w_2 \times 2 + ... + w_n \times x_n \tag{13}$$

  Where, $w_0, w_1, w_2, \ldots, w_n$ are coefficients representing feature importance.

  Interpret the coefficients: Examine these coefficients to understand the features' impact on the blending model's prediction for X_i. Positive coefficients indicate features that push the prediction towards class 1, while negative coefficients have the opposite effect. These coefficients provide insights into the local behavior of the blending model around X_i, explaining its specific prediction.

IV. EXPERIMENT AND RESULT ANALYSIS

A. EXPERIMENT SETTING

This section briefly overviews the dataset used and provides implementation details for the blending model.

We applied our proposed method to two datasets: CICIoT2023 [19] and IoTID20 [20]. In the first dataset, CICIoT2023, it comprises 47 input features and 8 output classes. To select the essential input features, we employed the MDI method. As depicted in Figure 2a, we identified the six most important features: 'IAT,' 'Magnitude,' 'Total Size,' 'Min,' 'Flow Duration,' and 'Total Sum.' The output classes include a 'Benign' class for normal behavior and seven attack class types, namely 'DoS,' 'DDoS,' 'Reconnaissance,' 'Web,' 'Brute Force,' 'Spoofing,' and 'Mirai.

In the second dataset, IoTID20, it encompasses 83 input features. After applying the MDI feature selection method, we pinpointed 11 of the most critical features, as illustrated in Figure 2b. These features consist of 'Timestamp,' 'Flow ID,' 'Flow Duration,' 'Bwd IAT Total,' 'Source Port,' 'Flow IAT Max,' 'Flow Packets per Second,' 'Flow IAT Min,' 'Initial Bwd Window Bytes,' 'SYN Flag Count,' and 'Category.' The output class of this dataset comprises five labels: 'Mirai,' 'DoS,' 'Scan,' 'Normal,' and 'MITM ARP Spoofing.'

The dataset was divided into training, validation, and testing sets following a 70:10:20 ratio. Besides, we utilized the Python programming language along with relevant libraries for model development. We establish the performance metrics employed to assess the efficacy of the suggested blending model. Standard metrics include accuracy, precision, recall, and F1-score. Additionally, we detail the training process, including the optimization algorithms, hyperparameter configurations, and cross-validation techniques employed in Table 2.

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TABLE 2. Hyperparameters setting for blending model.

Hyperparamters	Values
Learning rate	0.1
Loss	log loss
Max depth	3
Number of estimators	100
Splitter	123
Voting	soft

In soft voting, every base model computes the predicted probabilities for all classes, and the class with the greatest average probability, as determined by all models, is chosen as the ultimate prediction. Soft voting considers the confidence level associated with each model's prediction.

In binary classification, each base model computes predicted probabilities for both class 0 and class 1. The final prediction is made by selecting the class with the highest average probability across all models. This approach also factors the confidence level of each model's prediction for the two binary classes. Given: y_pred_GB: the Gradient Boosting model predicted probabilities for both classes (0 and 1). y_pred_DT: the Decision Tree model predicts probabilities for both classes (0 and 1). y_pred_RF: the Random Forest model predicts probabilities for both classes (0 and 1). We calculate the sum of predicted probabilities. For each class (0 and 1), calculate the sum of predicted probabilities across all three models. The sum of predicted probabilities for class 0:

$$sum_prob_0 = y_pred_GB_class_0 + y_pred_DT_class_0 + y_pred_RF_class_0 \tag{14}$$

The sum of predicted probabilities for class 1:

$$sum_prob_1 = y_pred_GB_class_1 + y_pred_DT_class_1 + y_pred_RF_class_1 \tag{15}$$

Where $y_pred_GB_class_0$, $y_pred_DT_class_0$, and $y_pred_RF_class_0$ represent the predicted probabilities for class 0 by the Gradient Boosting, Decision Tree, and Random Forest models, respectively. Similarly, $y_pred_GB_class_1$, $y_pred_DT_class_1$, and $y_pred_RF_class_1$ represent the predicted probabilities for class 1.

Each underlying model computes predicted probabilities for all individual classes in multi-class classification with soft voting. The ultimate prediction is determined by selecting the class with the highest average probability across all models. Given: y_pred_GB: Predicted probabilities by the Gradient Boosting model for all classes. y_pred_DT: Predicted probabilities by the Decision Tree model for all classes. y_pred_RF: Predicted probabilities by the Random Forest model for all classes.

To perform soft voting in multi-class classification, we calculate the sum of predicted probabilities for each class and select the class with the highest average probability. Here is how it works for multi-class: For each class (we assume there are n classes):

Step 1: Calculate the sum of predicted probabilities across all models for each class i:

$$sum_prob_i = y_pred_GB_class_i + y_pred_DT_class_i + y_pred_RF_class_i \tag{16}$$

Where: $y_pred_GB_class_i$ represents the predicted probabilities for class i by the Gradient Boosting model. $y_pred_DT_class_i$ represents the predicted probabilities for class i by the Decision Tree model. $y_pred_RF_class_i$ represents the predicted probabilities for class i by the Random Forest model.

Step 2: Calculate the average probability for each class i by dividing the sum by the number of models (m):

$$avg_prob_i = \frac{sum_prob_i}{m} \tag{17}$$

Step 3: Select the class i with the highest average probability as the final prediction:

$$final_prediction = argmax(avg_prob_i) \tag{18}$$

In this manner, the class with the greatest average predicted probability from all models is selected as the ultimate prediction in multi-class soft voting.

B. EXPERIMENTAL EVALUATION RESULTS

This section presents the experimental results evaluating our proposed ensemble blending model. We present various performance metrics obtained during the experiments, including accuracy, precision, recall, F1-score, AUC-ROC, training time, and prediction time measurements. To facilitate a clear understanding of our ensemble model's performance in detecting and explaining attacks in the IoT environment, we use tables and graphs to represent the results visually.

We illustrate the performance of our proposed method using learning curves to showcase the blending model's training and validation results. Additionally, we utilize confusion matrices to display the testing results, providing insights into detecting and explaining each attack type and normal behavior in multi-class classification.

- Model Training and Validation Results. We present the model training and validation results of our proposed method through learning curve plots for both datasets: CICIoT2023 (cf. Figure 3a) and IoTID20 (cf. Figure 3b). The overall results, as depicted in Figure 3, showcase the high-performance accuracy achieved by our proposed method on both the training and validation datasets.
- Model Testing Results. We present our validation results for the proposed method using testing data, as depicted in Figure 4. For the CICIoT2023 dataset, we illustrate the testing results in Figure 4a. These matrices display the performance of our model in classifying both 'Normal' and 'Attack' types. Notably, the 'Web-Based' attack type achieves a detection accuracy of 89%, primarily attributed to its smaller representation in the dataset compared to other attack types. This result highlights the effectiveness of our model in accurately identifying

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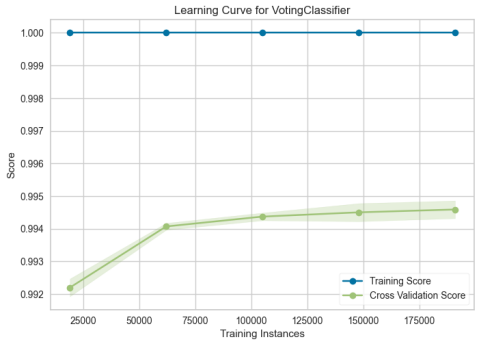
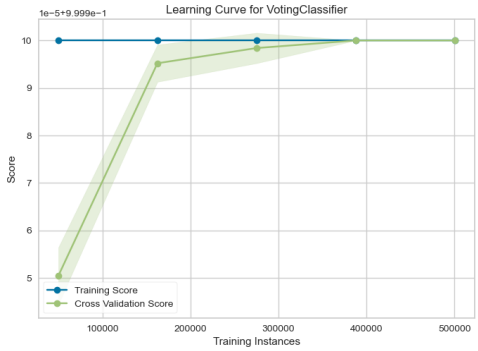
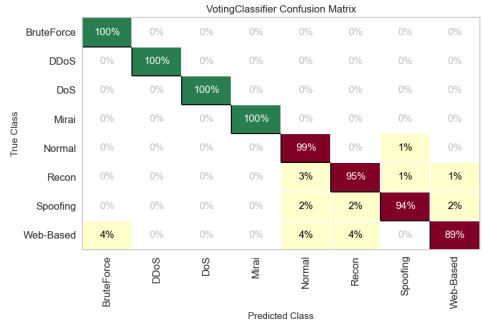


(a)

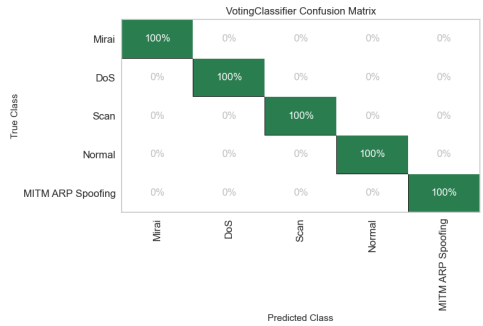


(b)

FIGURE 3. **Learning curve results of blending model. (a) CICIoT2023. (b) IoTID20.**



(a)



(b)

FIGURE 4. **Confusion matrix results of blending model. (a) CICIoT2023. (b) IoTID20.**

various attack types.

For the IoTID20 dataset, we provide the confusion matrix results in Figure 4b. Notably, our method demonstrates accurate detection of all attack types.

C. COMPARISON OF THE PROPOSED BLENDING MODEL WITH BASELINE METHODS

We conducted a comparative analysis to evaluate the performance of our proposed blending model against baseline methods employed for attack detection on the two datasets. We emphasize accuracy, precision, recall, and other pertinent metrics differences. This comparative analysis enables us to assess the superiority or advantages of our ensemble models over existing approaches and validates the effectiveness of our proposed ensemble models in enhancing attack detection and explanation.

Table 3 illustrates that our proposed method outperforms baseline models on the CICIoT2023 dataset within the same programming setting. Similarly, Table 4 demonstrates superior performance in attack detection on IoTID2020 compared to three baseline ML models regarding accuracy, AUC, precision, recall, and F1 scores.

In contrast, it is worth noting that each baseline model's training and testing times are shorter than the blending model. However, our blending model exhibits reasonable processing times, requiring 7 minutes and 4 seconds for training and testing on the CICIoT2023 dataset and 18 minutes and 7 seconds for training and testing on the IoTID20 dataset.

In addition, we conducted a comparison of our proposed method with another prior method on the same two datasets, as indicated in Table 5 for the CICIOT2023 dataset and Table 6 for the IoTID20 dataset. The results in both comparison tables demonstrate that our method outperforms other prior methods on the same datasets in terms of performance, including accuracy, AUC, precision, recall, and F1 scores.

D. LOCAL EXPLANATION FOR PREDICTION RESULTS

In this section, we analyze the interpretability and explainability aspects of the blending model, highlighting their ability to provide meaningful insights into the detected attacks. This analysis provides valuable insights into the effectiveness of the proposed ensemble blending model in the IoT environment.

- Decision Region Prediction. In our analysis, we employed the counterfactual method to visualize the decision region predictions generated by our proposed method. The results of these predictions are depicted in Figure 5. Specifically, Figure 5a showcases the outcomes of our blending method on the CICIoT2023 dataset, while Figure 5b presents the results for the IoTID20 dataset.

  A distinct pattern emerges after closely examining and comparing these counterfactual results. The decision region predictions generated by the counterfactual analysis of our blending model exhibit increased clarity and

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TABLE 3. Comparison classification of the proposed model with other single baseline models on the CICIoT2023 dataset.

Model	Accuracy	AUC	Precision	Recall	F1	Training time (s)	Testing time (s)
Extra Trees	0.9707	0.9898	0.9507	0.9408	0.9457	7.7872	0.6289
Random Forest	0.9536	0.9799	0.9536	0.9731	0.9633	10.7774	0.2839
Gradient Boosting	0.9640	0.9799	0.9440	0.9539	0.9489	159.5250	0.2700
Blending	**0.9951**	**1.0000**	**0.9851**	**0.9963**	**0.9907**	448.1058	3.8987

TABLE 4. Comparison classification of the proposed model with other single baseline models on the IoTID20 dataset.

Model	Accuracy	AUC	Precision	Recall	F1	Training time (s)	Testing time (s)
Extra Trees	0.9745	0.9808	0.9416	0.9628	0.9521	24.4911	1.0260
Random Forest	0.9812	0.9893	0.9656	0.9834	0.9744	42.3761	0.7699
Gradient Boosting	0.9534	0.9645	0.9532	0.9646	0.9589	339.5980	0.8360
Blending	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	1093.9912	7.0637

TABLE 5. Comparison classification of the proposed model with other prior methods on CICIoT dataset.

Model	Acc.	AUC	Pre.	Rec.	F1
Logistic Regression [19]	0.8316	-	0.5124	0.6960	0.5394
Perception [19]	0.8663	-	0.8663	0.8663	0.5551
Adaboost [19]	0.3513	-	0.4649	0.4877	0.3686
Random Forest [19]	0.9943	-	0.9943	0.9100	0.7192
Deep Neural Network [19]	0.9911	-	0.6794	0.9066	0.6972
Blending	**0.9951**	**1.0000**	**0.9851**	**0.9963**	**0.9907**

TABLE 6. Comparison classification of the proposed model with other prior methods on IoTID20 dataset.

Model	Acc.	AUC	Prec.	Rec.	F1
Ensemble [20]	0.87	-	-	-	-
CNN-LSTM [44]	0.98	-	-	-	-
RF [45]	0.9785	-	-	-	-
AutoEncoders [46]	0.945	0.912	-	-	0.971
SMOTE-SLFN-LSTM [47]	0.862	-	-	-	-
Extra Tree, RF, DNN [48]	0.9871	0.9825	0.9876	-	-
SVM-SMOTE-SLFN [49]	0.9351	-	0.9969	0.9578	-
DCNN [50]	0.9806	-	0.9726	0.9721	0.9723
Extra Tree, RF, DNN [48]	0.9871	0.9825	0.9876	-	-
PSO. XGB, RF [51]	0.83	-	-	-	-
Blending	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**

a reduction in the overlap between the classification regions of each attack class. This enhanced separation underscores the efficacy of our approach in achieving more distinct and precise decision boundaries for attack classification.

- Feature Important Distribution. We utilized the LIME method to visualize the feature importance that influences the decisions made by our blending model in two distinct scenarios: for correct and incorrect predictions. The results are in Figures 6 and 7.

  In the feature importance distribution for the CICIoT2023 dataset, as depicted in Figure 6a, it becomes evident that features with the most positive values, such as *IAT*, *Min*, *rst_count*, and *flow_duration* represented in blue, play a pivotal role in guiding the blending classifier to make accurate predictions. In contrast, Figure 6b reveals that features with the most negative values, including *IAT*, *Min*, *rst_count*, and *flow_duration* highlighted in red, significantly influence the blending classifier's tendency to make incorrect predictions.

  Similarly, in the case of the IoTID20 dataset, as shown in Figure 7a, accurate predictions by our method are influenced by features like *Timestamp*, *Flow_Pkts/s*, *Flow_Duration* indicated in blue. Conversely, the incorrect predictions, highlighted in red, are affected by features such as *Flow_Duration* and *Flow_IAT_Max*, and *Flow_Pkts/s*, contributing to erroneous decisions by our method.

- Correct and Incorrect Prediction Result Explanation. In Figures 8 and 9, we offer visualizations of LIME explanations for both correct and incorrect predictions generated by our blending method.

  To the CICIoT2023 dataset, for instances of correct classification, where the prediction aligns with the actual result ('Spoofing' attack), Figure 8a highlights the most influential features affecting the blending model's decision. Notably, features such as *IAT*, *ARP*, *Weight*, and *DNS* are identified as key contributors to the model's accurate prediction. Specifically, their values are recorded as follows: *IAT* = 83,331,790.15, *ARP* = 9.00, *Weight* = 141.55, and *DNS* = 0.00.

  Conversely, in cases of incorrect classification, where the prediction ('DDoS' attack) diverges from the actual result ('DoS' attack), Figure 8b illustrates the feature values that led to this misclassification. Features like *IAT*, *rst_count*, *flow_duration*, and *AGV* significantly influence the blending model's erroneous detection. Specifically, their values are recorded as follows: *IAT* = 0.00, *rst_count* = 5.60, *flow_duration* = 138.08, and *AGV* = 75.28.

  These LIME explanations shed light on the specific feature contributions that influence our blending model's correct and incorrect predictions.

  Similarly, for the IoTID20 dataset, we provide LIME local explanations for our method in two cases: correct and incorrect prediction classifications, illustrated in Figure 9. The explanation for correct predictions with a LIME result of 'Normal' value is shown in Figure 9a, and the explanation for incorrect predictions with a LIME result of 'DoS' as the prediction value while the actual label is 'MITM ARP Spoofing' is displayed in Figure 9b.

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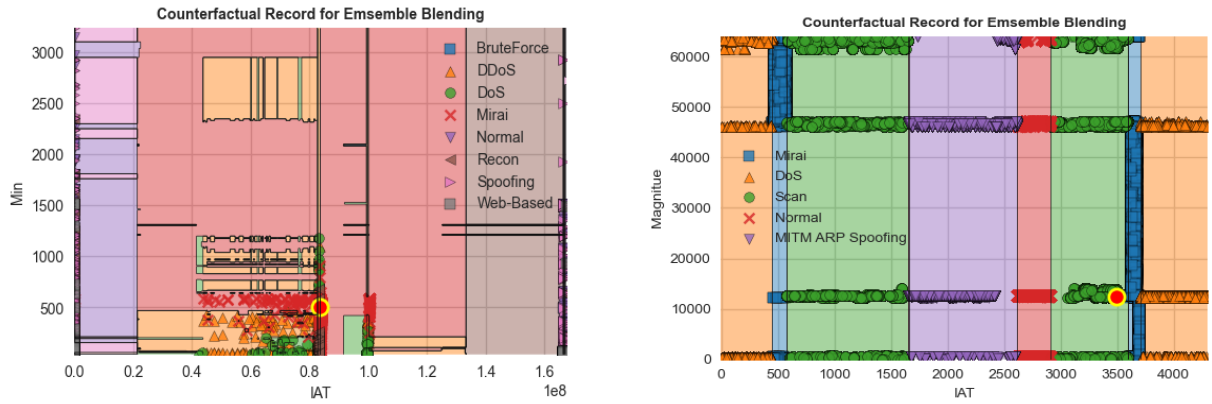

(a) (b)

FIGURE 5. Counterfactual record visualization of decision region prediction on two datasets. (a) CICIoT2023. (b) IoTID20

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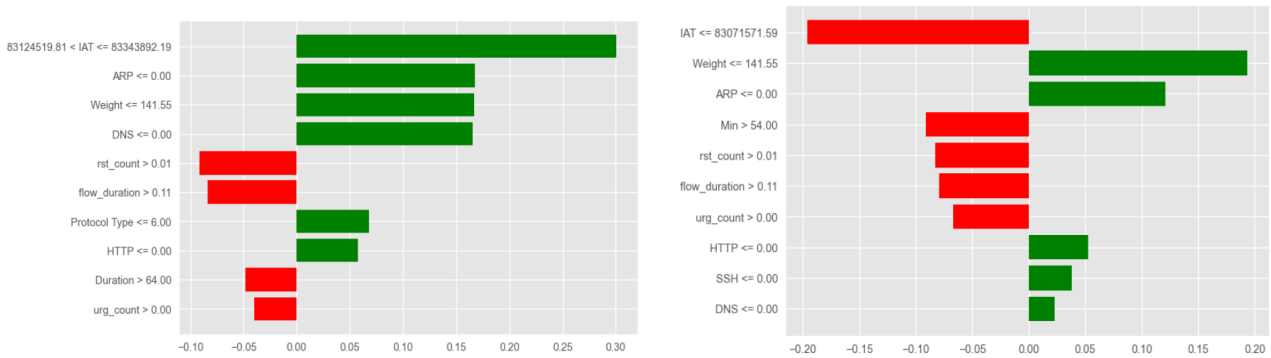
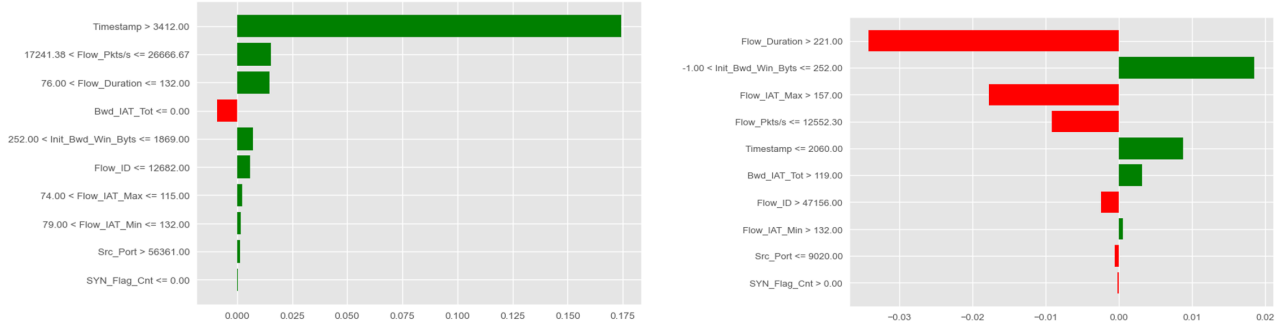


(a) Correct prediction case (b) Incorrect prediction case

FIGURE 6. Local explanation with feature contribution explanation for blending classification model in two cases: correct and incorrect classification on CICIoT2023 dataset. (a) Classification value: Spoofing, actual value: Spoofing. (b) Classification value: DDoS, actual value: DoS.



(a) Correct prediction case (b) Incorrect prediction case

FIGURE 7. Local explanation with feature contribution explanation for blending classification model in two cases: correct and incorrect classification on IoTID20 dataset. (a) Classification value: Normal, actual value: Normal. (b) Classification value: DoS, actual value: MITM ARP Spoofing.

V. DISCUSSION

The discussion section involves the interpretation of the results and their implications, the advantages and limitations of the proposed approach, considerations and challenges for deployment in real-world IoT environments, and exploring potential applications and future research directions. This comprehensive discussion provides a holistic view of the proposed approach and its broader implications for improving attack detection and explanation in the IoT environment.

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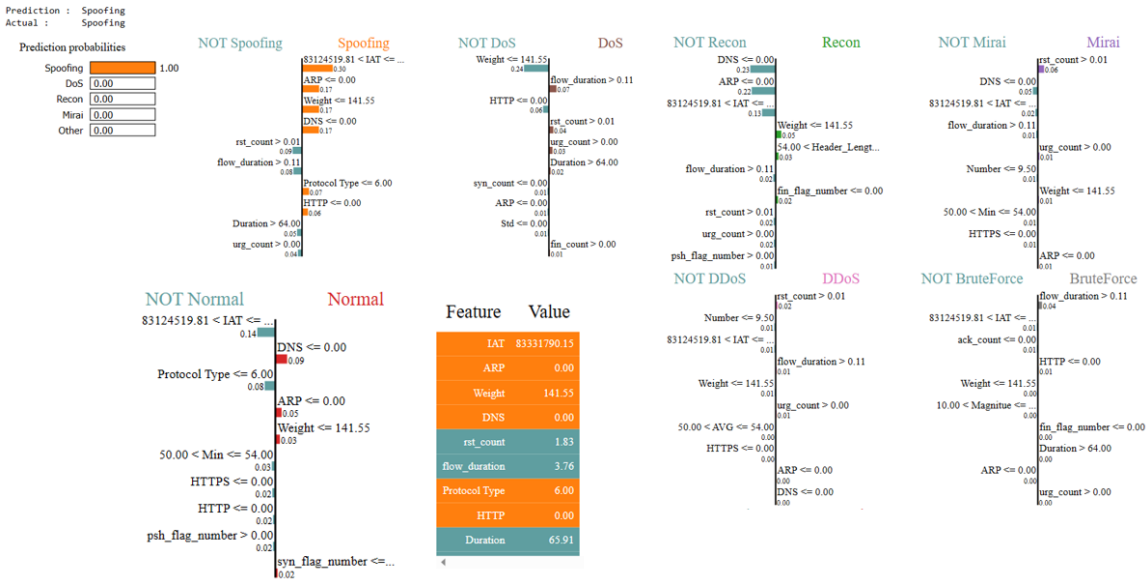
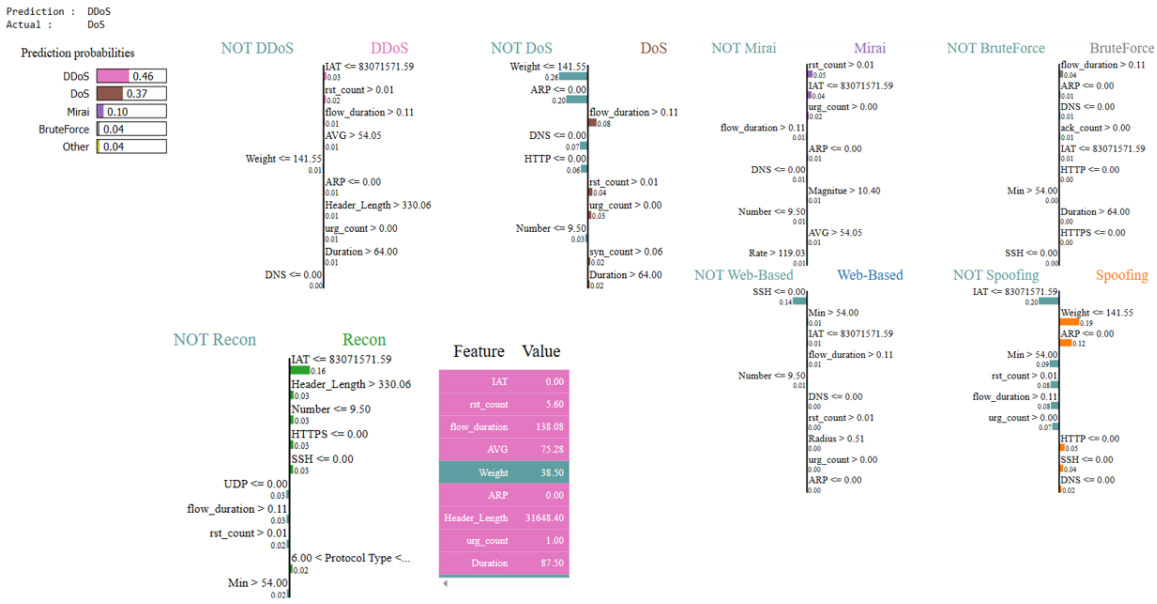


(a) Correct classification result



(b) Incorrect classification result

FIGURE 8. Local explanation for blending classification model in two cases: correct and incorrect classification on CICIoT2023 dataset. (a) Classification value: Spoofing, actual value: Spoofing. (b) Classification value: DDoS, actual value: DoS.

A. THE ADVANTAGES AND LIMITATIONS OF THE PROPOSED APPROACH

In this section, we delve into the results obtained from our experiments, dissecting their implications and providing a comprehensive analysis of the performance metrics achieved by our proposed ensemble blending model. We interpret these metrics in the context of attack detection and explanation within the dynamic IoT environment. This examination is crucial for gaining a deeper understanding of the effectiveness of our approach.

Our ensemble blending model has demonstrated several key advantages:

- Improved Accuracy: One of the standout features of our approach is the substantial improvement in attack detection accuracy compared to baseline models. By combining the predictive power of multiple models, we achieve a higher level of precision in distinguishing between benign and malicious activities in the IoT environment.
- Enhanced Interpretability: The integration of counterfactual and LIME explanation techniques has significantly enhanced the interpretability of our model. Security analysts and stakeholders can gain transparent

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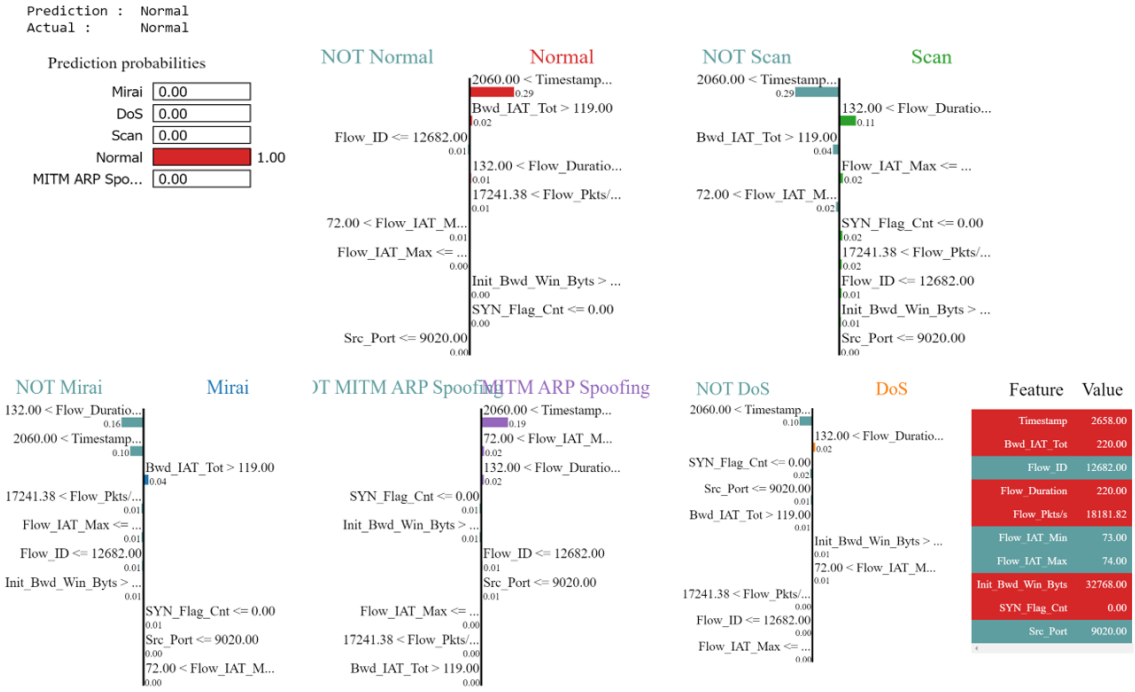
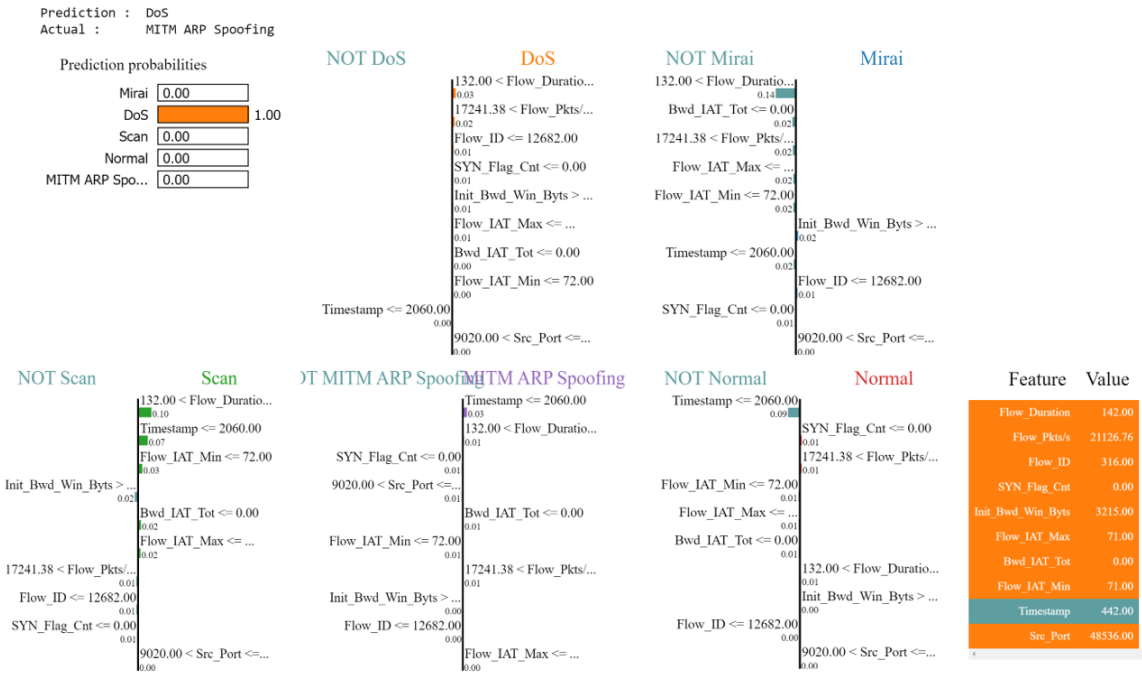


(a) Correct classification result



(b) Incorrect classification result

FIGURE 9. Local explanation for blending classification model in two cases: correct and incorrect classification on IoTID20 dataset. (a) Classification value: Normal, actual value: Normal. (b) Classification value: DoS, actual value: MITM ARP Spoofing.

insights into the factors driving classification decisions, facilitating more informed decision-making.

- Collective Intelligence: The ensemble blending model capitalizes on the collective intelligence of multiple models with distinct characteristics. This collective wisdom allows for better generalization, increased robustness, and effective addressing diverse attack scenarios.

While our ensemble blending model offers promising advantages, we must also acknowledge certain limitations and challenges:

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- Model Selection: The effectiveness of ensemble models is contingent on selecting appropriate base models. Identifying the right combination of models and their configurations remains a non-trivial task and requires careful consideration.
- Data Availability: Our model uses labeled training data to learn and make predictions. In real-world IoT environments, obtaining large-scale labeled datasets can be challenging, potentially limiting the model's applicability.
- Scalability: Although our approach has shown effectiveness in the experimental setting, scaling it to handle large-scale IoT environments with many devices and data streams may require further optimization and resource allocation.
- Resource Requirements: The computational and memory resources required for training and deploying ensemble models can be substantial. Ensuring that the model remains efficient in resource-constrained IoT environments is a consideration for practical implementation.
- Generalizability: The generalizability of our model to various IoT ecosystems, each with its unique characteristics and challenges, warrants further investigation. Achieving robust performance across diverse IoT deployments is a complex task.

This discussion provides a balanced assessment of our ensemble blending model's strengths and weaknesses. While we celebrate the significant improvements in accuracy and interpretability, we remain mindful of the practical challenges and considerations accompanying its implementation in real-world IoT security scenarios. By acknowledging these limitations, we pave the way for future research and refinement of our approach, ultimately contributing to more secure IoT environments.

B. PRACTICAL CONSIDERATIONS AND DEPLOYMENT CHALLENGES IN REAL-WORLD IOT ENVIRONMENTS

This section delves into the practical considerations and deployment challenges of implementing our proposed ensemble blending model within real-world IoT environments. We discuss the feasibility of deploying these models at scale in large IoT networks, considering critical factors such as resource constraints, data privacy, and computational requirements. Furthermore, we consider the challenges inherent in collecting and preprocessing IoT data, model maintenance, and the need for continuous model updates. Understanding these practical considerations and deployment challenges is pivotal for translating our research findings into tangible, real-world applications.

1) Feasibility in large-scale IoT networks: Deploying an ensemble blending model in large-scale IoT networks raises practical questions and challenges. We must consider resource limitations regarding computational power and memory, as many IoT devices operate with constrained resources. Adapting our model to function effectively within these constraints is crucial for practical deployment.
2) Data privacy and security: IoT environments handle sensitive data, making data privacy and security paramount. As we extend our model's deployment, ensuring the confidentiality and integrity of data becomes a significant concern. Strategies for data anonymization, encryption, and secure communication must be integrated into our deployment framework.
3) Data collection and preprocessing: Effective deployment hinges on the quality and availability of training data. Collecting and preprocessing IoT data can be complex, with challenges related to data diversity, missing values, and data drift over time. Addressing these challenges is essential to maintain model accuracy and robustness.
4) Model maintenance and updates: IoT environments are dynamic, with evolving attack vectors and device configurations. Ensuring our model remains effective over time requires a strategy for continuous model monitoring, maintenance, and updates. It includes retraining the model with fresh data to adapt to emerging threats.
5) Expanding applications: Beyond attack detection and explanation, we explore the versatility of our ensemble blending model. It can potentially find applications in various security domains beyond IoT, such as network security, industrial control systems, and smart cities. Additionally, we consider its adaptability to diverse IoT devices and communication protocols.
6) Future research directions: Identifying future research directions is integral to advancing our proposed approach. Areas of exploration may include: (1) Investigating novel ensemble techniques to enhance model performance and efficiency; (2) Exploring the integration of deep learning models for more sophisticated threat detection; (3) Developing real-time monitoring and response mechanisms to augment our model's proactive security capabilities.

By identifying potential applications and research directions, we aim to contribute to the continuous evolution of the field, fostering innovation and ultimately enhancing the security of IoT environments.

VI. CONCLUSION

This paper has emphasized enhancing attack detection and explanation within IoT environments through the innovative application of the ensemble blending model. These models harness the collective intelligence of multiple algorithms, leading to a substantial improvement in attack detection accuracy and the provision of interpretability. Rigorous experiments using two representative datasets have reaffirmed the efficacy of the proposed models.

The contributions of the ensemble blending model to IoT security are noteworthy. They offer heightened accuracy and a deeper understanding of attack classifications. By addressing the shortcomings of existing approaches, these models

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provide a more robust solution for attack detection and explanation. It, in turn, has significant implications, including improved safeguarding of IoT devices and networks, swift response to attacks, and formulating comprehensive security strategies. Ultimately, the proposed models create a safer and more dependable IoT ecosystem.

In conclusion, the ensemble blending model introduced in this paper represents a promising avenue for elevating attack detection and explanation in IoT environments. Continued research and development in this realm will undoubtedly propel the field of IoT security, fortifying the foundations for a more secure and resilient IoT landscape.

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